



SahAI

AI Co-Pilot for India's India's 1.3M ASHA Workers

"Augmenting judgment. Not replacing it."

AI4INDIA HACKATHON 2026

Meet the Team

A multidisciplinary team built for real-world health impact — from model to mission.



ML Engineer

On-device NLP, ASR
quantization, TFLite risk models



Public Health Researcher

NHM protocols, ASHA
Workflow design, field
validation



Product Designer

Voice-first UX, low-literacy
interface, field usability



Backend Developer

Sync layer, FHIR APIs,
anonymization pipeline

The Problem

1.3M

ASHA Workers

India's last-mile healthcare backbone

97

MMR per 100K

Maternal deaths still preventable

1:10K

Doctor Ratio

Rural India's crushing healthcare gap

57%

Time on Paperwork

Not on care. On compliance.

"An ASHA worker visits a pregnant woman with dangerously high BP. No digital checklist. No risk protocol. Paper records. Escalation comes too late."

❏ This is not a system failure. It is an information failure.



SahAI

Sahayak + AI — "Your AI Assistant"

An offline-first voice AI co-pilot built for India's frontline health workers. No internet. No literacy barrier. No friction.

- Voice input in local languages
- Auto-converts to structured health summaries
- Flags risks using rules + ML engine
- Generates follow-up plans instantly
- Runs on low-end Android, zero connectivity needed

Primary Impact

1.3M ASHA Workers

Secondary Impact

600M Rural Citizens

System Impact

NHM · ANMs · PHC Doctors

Field Personas

Designed from the ground up for the humans who will use it every day.



Savitri Devi — ASHA Worker

Goals: Serve her community, complete visits efficiently, feel confident in high-risk cases

Pain Points: Overwhelming paperwork, no real-time guidance, fear of missing danger signs



Rekha Sharma — ANM ANM Supervisor

Goals: Monitor her ASHA cohort, catch risks early, reduce emergency escalations

Pain Points: Fragmented paper reports, no real-time visibility, delayed referral information



Arvind Nair — District Manager

Goals: Population-level insights, NHM compliance, measurable outcomes

Pain Points: No aggregated signals, reactive decisions, no early warning system

The Solution

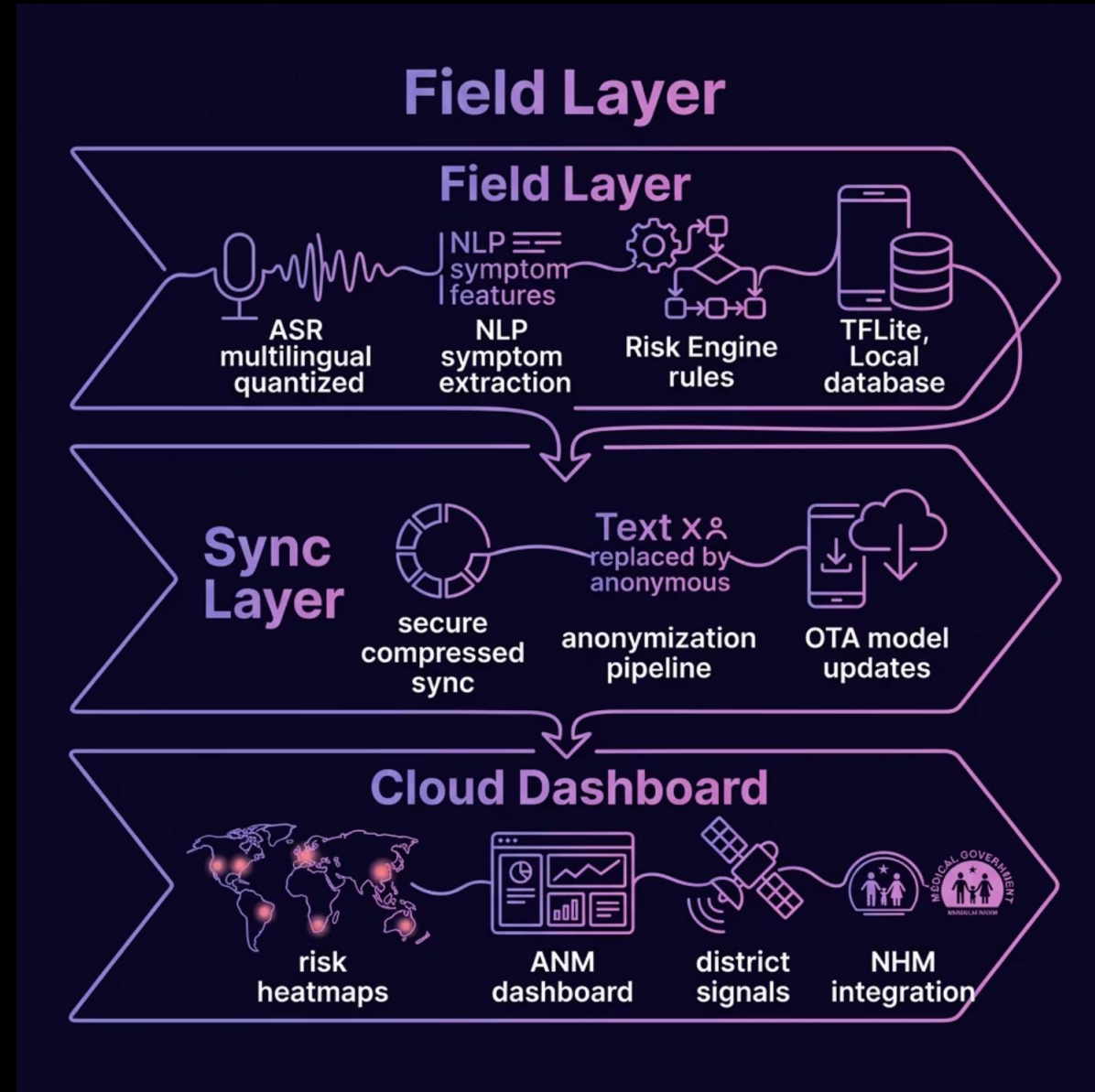
How SahAI works in the field — from voice to action in seconds



❏ Referral delay transformed: 14 days → Same day — the single most critical improvement in maternal outcomes.

Architecture

Three integrated layers — built for resilience at the edge of connectivity.



Field Layer

Fully offline. ASR + NLP + Risk Engine runs entirely on-device using TFLite.

Sync Layer

Compressed, anonymized batch sync when connectivity is available. OTA model updates.

Cloud Dashboard

Risk heatmaps, ANM oversight, district-level signals, and NHM API integration.



Cognitive Augmentation, Not Automation



Multilingual Voice Intelligence

Hindi, Tamil, Bengali + 8 regional languages. Quantized ASR for low-end devices.



Hybrid Risk Scoring

Clinical rules + TFLite ML model. Transparent, auditable, always explainable.



Protocol-Aware Content

MOHFW and NHM guidelines embedded. Every recommendation is policy-grounded.



Community Signal Aggregation

Individual visits → district-level patterns. Epidemiological intelligence from the ground up.



SahAI never diagnoses. It supports decisions. The ASHA worker remains the trusted human in the loop — always.

Challenges & Readiness

Every risk anticipated. Every mitigation built in from day one.



Literacy & Adoption

Voice-first UI with no reading required. Onboarding in under 10 minutes.



ASR Noise Robustness

Confidence scoring with in-app correction prompts for ambiguous speech.



Data Privacy

On-device processing + anonymization before any sync. DPDP Act compliant.



Govt Integration

FHIR-compliant APIs ready for NHM, HMIS, and ABDM interoperability.



Model Drift

OTA updates + continuous expert validation loop ensure clinical accuracy over time.

Thank You

60%

Docs Time Saved

Documentation burden cut dramatically

14→1

Days to Decision

Referral delays eliminated

20%

Preventable Deaths

Projected reduction in mortality

500M+

Annual Data Points

Community health intelligence generated

"SahAI doesn't replace the ASHA worker. It makes her unignorable."

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